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IFC-centric performance-based evaluation of building evacuations using fire dynamics simulation and agent-based modeling

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ABSTRACT

Despite recent technological advancements in the architecture, engineering, and construction (AEC) industries, the number of fire-caused fatalities in residential and commercial buildings in the United States over the last decade has consistently been an order of magnitude higher than fatalities caused by all other types of natural disasters combined. In this research, EvacuSafe is developed as a tool for designers to allow for layout optimization of buildings and facilities in terms of their evacuation safety performance. Contributions of this research include the development and validation of two spatiotemporal risk indices to quantify the safety of the egress routes and individual building compartments. Other unique advantages of EvacuSafe include the seamless integration of fire dynamics simulation, agent-based crowd simulation and building information models (BIM) using IFC data structures, allowing EvacuSafe to be readily used by designers to analyze a building layout design under various fire scenarios and for layout design optimization based on multiple safety criteria. The development of the framework, the risk indices, and the IFC-based integration is illustrated and validated through a case study on a two-story office building with 59 compartments. The results of the implementation show that the EvacuSafe is a valuable tool for evacuation design and planning that provides a more comprehensive evaluation of the evacuation performance in comparison to the existing indices and safety measures in the industry.

1. Introduction

In 2015, 2635 fire-caused deaths occurred in residential and non-residential buildings in the United States [1,2]. This number is consistent with the annual average of 2675 fire deaths that occurred during 2006–2010, although it disregards normalizing factors such as population growth. The lack of improvements in this area is counterintuitive given the tremendous scientific and technological advancements that have been accomplished during the same period. Compared to 317 fatalities in 2015 caused by all other types of natural disasters in the United States, fire losses are an order of magnitude higher. This demonstrates the need for further constructive considerations in this area [3].

Since fire extinguishment solutions cannot be successful in all fire cases, fire safety strategies and regulations are typically based on the timely evacuation of the building [4]. Occupants are prone to being trapped during the evacuation if they do not have or are not aware of the emergency plan. In a review of fire fatalities in residential buildings between 2013 and 2015, 19.8% were caused by egress problems and 16.8% were caused by escape problems [5]. Egress problems refer to crowded situations, limited exits, locked exits, and mechanical obstacles to the exit. Escape factors refer to unfamiliarity with exits, excessive escape distance, inappropriate choice of exit route, re-entering the building, and clothing catching on fire while escaping [5]. Although evacuees could lose their lives due to a failed emergency plan, a studious design, which contemplates the safe and the fast egress of the occupants, can significantly lower the risk.

Evacuation research primarily focuses on reducing travel time or travel distance of evacuation through the selection of the best route or development of the optimal design. However, a challenging question still exists: "to what extent can an evacuation plan that is solely evaluated on evacuation time guarantee safe egress from a building?". Fire cases differ in terms of their location in the building, the source of fire, functionality of the building, and means of egress. It may be impractical or simplistic to measure the performance of evacuation, whose main goal is to save lives, by a single number representing the time required to vacate the entire building. An evacuation could take longer but result in fewer casualties or fatalities. Moreover, the time/distance-based approach looks at the building without considering its individual compartments and routes. For instance, a specific part of the building could be at immediate risk because of its proximity to the fire, while the rest of the building could be relatively safe. The evacuation time cannot

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transfer this semantic information or provide a basis for comparison of different fire scenarios.

Therefore, a detailed assessment of evacuation processes can be achieved through simulating fire and occupant behavior, and the creation of indices that report risk levels in different zones of the building. For this purpose, the main objectives in this paper are to:

1. Define a risk index to quantify the safety of the egress routes and compartments in fire emergency cases; and,
2. Develop EvacuSafe, a framework for evaluating the evacuation safety performance of buildings based on the developed risk indices and through integrating agent-based crowd simulation techniques with fire simulation tools and building information models.

2. Literature review

2.1. Crowd simulation approaches

Crowd simulation approaches could reflect individual and crowd decision making during evacuation processes. An ideal simulation can model the occupants' behavior, building characteristics, and the dynamic nature of the hazard. Flow-based models, cellular automata, social force models (SFM) and agent-based models are of the most widely used approaches of crowd simulation.

Flow-based modeling uses the principles of fluid and particle motions to simulate crowds. The graphical interface consists of nodes and arcs, where nodes represent the areas containing people with a calculated useable area and arcs represent the flow of passageways between building components per unit time. Flow-based software such as EVACNET4, EESCAPE, and EGRESSPRO, have no provisions for modeling the social behavior or individuals' actions during an evacuation [6–8]. Further, they do not have the ability to perform a simultaneous fire modeling in the crowd simulation environment [9].

The cellular automata approach plots the modeling space into a two-dimensional matrix. Entities can move from their current node to one of 8 adjacent nodes if it is not occupied by another entity. This system has limitations in terms of its ability to represent real-life movements and modeling entities that move at different speeds [10]. Cellular automata models are discrete in time and space. Since they typically use square meshes, they are subject to error because of incorrect choice of map projection or directional biases [11]. Behavior during emergencies [12], herding [13], and competitive egress behaviors [14] have been studied using the cellular automata approach.

SFM is one of the most widely used approaches in pedestrian simulation modeling. Here, individual pedestrian behaviors are subject to external forces. These forces could be a result of their interactions with other pedestrians, their physical environment, or their motivation to act. In the initial version of the social force model [15], several effects on the motion of the pedestrians were considered, including their desire to achieve their destination, avoid or approach objects and other pedestrians, and sidestep borders and obstacles. The resultant force defines the state change of the pedestrians over time. The model was later improved and included the effect of panic in simulation [16]. SFM have been applied to simulate the evacuation from a room with one exit [17,18], the effect of leadership within multi-exit evacuations [19], and the effectiveness of wall versus ground exit signs [81].

ABM is an approach to simulating the actions and interactions of autonomous agents in a modelled environment [20]. The ability of ABM to model populations, individual behaviors, and agent interactions make it very suitable for crowd simulation. ABM's bottom-up approach allows the model to imitate the unique behaviors of heterogeneous humans, including emergent behaviors of panic, competitiveness, queuing, herding, and the kin effect [12,21–23]. In ABM, the user chooses the most appropriate human behavior factors to influence the physical movements of the agents inside the system. The main challenge is to understand these emergent human behaviors and to translate them into mathematical rules. ABM has been combined with SFM to simulate crowd evacuation in emergency events, such as a terrorist attack or earthquake [24–26].

Building geometry and fire scene elements have been incorporated into ABM to simulate the interactions between the agents and the environment during an evacuation [27]. Fire Dynamics Simulator (FDS) is a software based on computational fluid dynamics models that models the migration of fire, heat, and smoke [28]. FDS and graphic information system (GIS) data were integrated to provide fire scene and building geometry in agent-based evacuation models [27]. FDS + EVAC, an agent-based fire evacuation model, showed how the effects of a fire, i.e. temperature and poisonous gases, impacted the movement of agents [29].

In summary, crowd simulation has been complemented by incorporating human emergent behaviors under emergency conditions, such as crowd panic when it encounters fire [30]. In this context, ABM enables a fire scene to be embedded as a continuous agent in the modeling environment and to interact with human agents and building physical properties.

2.2. Evaluation of building evacuation process

Several measures exist to assess and report the quality of an evacuation plan. Evacuation time was first introduced in 1917 as the "maximum emptying time" [31]. The term evolved and was eventually defined as the time between the activation of an alarm and the completion of the evacuation [32]. Parameters such as average evacuation time, average travel distance, individual exit flow rates, and average waiting time have also been used [33]. Evaluation based on these concepts ignores the delay caused by hazard recognition and response times of individuals, and information related to the dynamic nature of a hazard.

In 1983, two new concepts were introduced in fire safety engineering [34]. Required safe egress time (RSET) is the time for occupants to get out of a building safely. RSET has been interpreted in different ways; however, the most comprehensive definition starts at the moment of fire ignition and includes the time for detection, alarm activation, occupant decision to take action, and the travel time for the occupant to reach a safe place [35]. Available safe egress time (ASET) is the time it takes for a fire to make the building untenable. It is defined as the time from fire alarm activation to when the upper level gas layer of the fire descends to the level of an occupant's head [32].

The objective in this approach is to keep ASET > RSET. The difference between ASET and RSET is the safety margin (t_{ma}), where $t_{ma} > 0$ for a safe evacuation. A safety index (SI) to benchmark the life safety of occupants [36] was defined as the ratio of t_{ma} to RSET:

$$SI = \frac{t_{ma}}{RSET} \tag{1}$$

RSET/ASET has been adopted by many national organizations, including National Research Council of Canada (NRCC), International Organization of Standardization (ISO), and British Standard Institution (BSI) [37–41]. Although these organizations and standards have promoted the application of RSET and ASET, their prescriptive codes barely provide a clear procedure for the quantification and calculation of RSET/ASET [39,42]. For instance, recommended pre-movement time in the calculation of RSET ranges from < 80 s to 11 min [41,43]. Likewise, different levels of visibility, obstruction, poisonous gas concentrations, oxygen availability, and temperature have been suggested to define tenability thresholds [44]. These varied levels are linked to subjective life safety tolerances and explain the lack of objectivity on the computation of RSET/ASET.

In RSET/ASET, the RSET is treated as a deterministic number whose comparison with ASET determines the safety or lack of safety of a scenario. Considering the probabilistic nature of the influencing factors, such as congestion, occupant speed, human behavior, and fire expansion patterns, RSET should vary according to building-specific characteristics and events. A single number cannot represent all possible scenarios. Another flaw in RSET/ASET is that the location and approaching direction of the hazard is not taken to account. For instance, two fire scenarios could have similar ASET and RSET; however, in one case, more evacuees might encounter the untenable location during evacuation from the building. This misperception can happen between two cases of fire in a 30-story building, one in the first floor close to exit door and one in 29th floor.

2.3. Fire simulation

Mathematical fire modeling has been used since the 1940's in the design of fire protection systems, estimation of fire risks, and study of the effects of fire on people and properties [45]. Mathematical fire modeling is categorized as 1) hand calculations, 2) zone models, and 3) computational fluid dynamics (CFD), also known as field models [46]. Hand calculations are a collection of algebraic equations inferenced from empirical experiments to predict the characteristics of a specific fire phenomenon. Zone models divide the compartment to an upper zone and a lower zone and investigate the mass and energy transfer between these two spaces. The upper zone consists of the fire plume and gas products while the lower zone contains the ambient air or non-smoke layer. As time passes, the separation between these two layers moves because of mass transfer. Consolidated Model of Fire Growth and Smoke Transport (CFAST) developed by U.S. National Institute of Standards and Technology (NIST) is the most common fire modeling software and is based on zone modeling [47]. Field models, which are relatively more computationally intensive, divide the space into thousands of cubic cells and study energy and mass transfer between them. FDS developed by NIST is a well-known computer program of field model types [28].

Path planning and evacuation modeling without considering the effect of the fire scene may give results that are misleading as it disregards the physical attributes of the environment. In other words, an evacuation model can be significantly improved when path planning, fire simulation, and crowd simulation modules dynamically interact. In real-life situations, fire expansion patterns continuously change the results of path planning and affect the movement and behavior of the occupants. CUrisk, developed at Carleton University, attempts to combine sub-models of smoke and fire growth, occupant evacuation and property economic loss to perform risk assessment on fire safety designs [48]. In CUrisk a 2-zone model is used as the fire simulator. Life safety is quantified through an expected risk to life (ERL) measure. ERL is the expected life loss during (ELLD) the design life of a building divided by the building's population and the expected design life. ELLD is calculated as the number of simulated deaths in each fire scenario and probability of each scenario occurring during its design life [49,50]. CESARE-Risk, FiRECAM [51], and FRAMEworks [52] are risk assessment models based on ERL that combine fire simulation and evacuation modeling to quantify the number of expected deaths in each scenario. The focus of the aforementioned models is not specifically on evacuation planning and compartment risk assessment, but rather on fire risk assessment of the building and its occupants.

2.4. Building information modeling (BIM) & industry foundation classes (IFC)

BIM is a 3-dimensional digital representation of the physical and functional characteristics of a facility [53]. BIM data structures are such that they could be queried both through visual and numerical methods.

BIM's capabilities have been explored for a range of emergency management applications, such as assessment of building safety features before an emergency, immediate access to building information for evacuation modeling, and training simulators for emergencies and post-event assessment of emergencies [54–57]. Recent research initiatives in this area show an increasing impetus towards the application of BIM for fire safety management. A framework was designed to help rescuers find the shortest path within a fire scene to the triggered fire detector [58]. The model receives data related to the building 3D objects from the BIM model and uses wireless local area network (WLAN), ultra-wide-band (UWB) and radio frequency identification (RFID) for indoor navigation. In another research, a BIM model was used to acquire the inputs for FDS, and then ASET was computed by running the FDS model [59]. RSET was calculated based on local building safety regulations. ASET and RSET indices were then compared to examine the safety level of the building in emergency cases. BIM models have also been combined with serious game environments to study occupant's behavior and responses in fire scenarios [60]. Using this method, there is no need to build every single game scenario from scratch and the BIM model can be used to transfer the building conditions efficiently. Moreover, BIM data structure has been used for automated code checking in evacuation design [61]. Automated code checking for evacuation and fire safety design can significantly improve the efficiency and accuracy of regulation compliance in iterative process of architectural design and permit acquisition.

Currently, many software applications in architecture, engineering and construction (AEC) use BIM technology to improve the efficiency of design, planning, construction and management of buildings. Data exchange and interoperability between such software applications required a standard for transferable data structures. IFC was created as a data format to facilitate interoperability between AEC software applications. IFC is frequently used as a data format that enables the collaboration between workspaces dealing with BIM-based projects. In the context of indoor navigation, IFC geometric and semantic data have been used to extract the navigable locations of indoor 3D spaces [62]. To annotate the nodes of a 2D grid mesh, IFC elements were tagged as navigable or obstacle (unnavigable) with 1/0 values. Semantic information from the IFC was added to the grid representation as well. For example, locked doors generated unnavigable nodes in the grid mesh. Also, a risk factor was proposed around distribution lines, so that the occupant would be pushed away from high-risk zones. Multi-Purposed Geometric Network Models (MGNM) derived from IFC were another method to build the network graph of the 3D indoor spaces by connecting each space to its surrounding elements [63]. In the generated network model, the nodes are spaces and portals (e.g. doors), and the edges are the routes between the nodes. The resulting network model, updated with geometric information of the IFC elements, provided an indoor navigation model for emergency response and path planning. On the other hand, some researches tried to transfer the geometric and semantic information of BIM models to fire simulation modules. IfcSTEP-FDS was a software application to parse the IFC file and convert it to an input file readable by FDS [64]. As another example, CYPECAD MEP software that has a visual fire simulator module developed based on FDS and SMOKEVIEW, allows the import of IFC files for developing fire simulations [65]. Moreover, IFC data can be used to generate the modeling environment in crowd simulation engines [66]. This data exchange significantly enhances the efficiency and accuracy of modeling process. Overall, it is evident that the concept of IFC data structures has great advancements in bridging software applications between the disciplines of the AEC industry.

3. Model development

The objectives of this research include the development of EvacuSafe as a framework for evaluating the evacuation safety performance of buildings based on two multi-criteria risk indices and through the integration of ABM, fire simulation tools, and building information models.

In this section, two risk indices are defined to quantify the safety of the egress routes and compartments in fire emergency cases. The defined indices are used as the basis for the development of the framework of EvacuSafe. Implementation of fire simulation, path identification and crowd simulation modules and their interconnections are also elaborated.

3.1. Route & compartment risk indices

Emergency response plans in the form of evacuations normally have quantifiable measures to determine the success of an evacuation. The problem with the time and the distance measures is that they do not provide knowledge about the risk of using a specific egress route or of being in a specific compartment in the building. Therefore, there is a need for a measure that:

- Considers the proximity of the occupant to the hazard during evacuation
- Traces the safety state of the occupant during evacuation
- Depends on spatiotemporal status of the occupant

In this research, Available Local Egress Time (ALET) is introduced. ALET is the time available for each point in the building, at each instant of time, until it will become untenable. The tenability criteria may differ based on the desired margin of safety. Lower ALET (i.e. closer to the fire) means less time available until un-tenability criteria will be met in that location. Therefore, the ratio of 1/ALET can be a measure for the risk of being present at any location in the building. As the evacuee moves during the egress process, their 1/ALET changes along the egress path. This measure needs to be treated as a time-dependent variable. In other words, staying longer on a point even with a fixed 1/ALET ratio measure has to escalate the risk index. In this way, the length of exposure to a specific proximity to fire is taken into account. Therefore, based on the requirements of the proposed framework, the following formulas have been established.

$$RRI_{j,k} = \int_{t=0}^{T} \frac{1}{ALET_{p(t)}} \mathrm{d}t \tag{2}$$

$$CRI_j = \frac{\sum_{k=1}^{K} RRI_{j,k}}{K} \tag{3}$$

where:$\mathrm{ALET}_{\mathrm{p(t)}}$ is the time it takes for a defined fire to reach $\mathrm{p}_{(\mathrm{t})}$ (point p at time t)CRI_j is the Compartment Risk Index of jth compartmentK is the total number of possible egress routes for compartment jp(t) is the location (x,y,z) of a defined evacuee at time t on the kth egress route$\mathrm{RRI}_{\mathrm{j,k}}$ is the Route Risk Index related to the kth possible egress route of compartment jT is total travel time of the defined evacuee from starting point to the end of egress route.

In this framework, a compartment is a separate navigable space in which certain elements can be kept separate from other spaces. This separation could be in terms of access, boundary elements, functionality etc. In this definition, rooms, corridors, lobbies, elevator cabins and stairwell landing areas are compartments.

These two indices, formulated in Eqs. (2) and (3), support the analysis of a design for safe evacuation. The analysis objectives are:

- To detect blocked egress routes (i.e. routes that contain at least one untenable location)
- To detect blocked compartments (i.e. compartments whose egress routes are all blocked)
- To find the safest egress route from each compartment
- To determine the most critical location for initiation of fire
- To determine the critical zones in the design (i.e. that have the highest risk in different fire scenarios)

In RRI formula, $t = 0$ represents the moment that the fire is initiated. From the design prospective, the initiation of fire is a user-defined element and the designer decides when to trigger the simulation model. Therefore, sensors and detectors will be activated at a time greater than $t = 0$. In addition, ALET is calculated right after fire initiation and it does not matter whether the occupant is moving or not. Herein, it is assumed that the existence of fire is confirmed, evacuation is mandatory and there is no false alarm in the building.

Given Eq. (2), a larger travel time or being closer to the fire (ALET) will lead to a larger area under the curve of 1/ALET vs. elapsed time; and consequently a larger RRI. The goal is to lower RRI for a higher level of safety. The reciprocal function of ALET will measure the proximity to the fire and integration over travel time will incorporate the travel time. Since it is impractical to measure 1/ALET for every point on an egress route, the space can be divided into sections or compartments as discrete points where ALET can be examined at the compartment center points. Where the compartment is untenable, i.e. ALET = 0, RRI becomes infinity, indicating a blocked egress route. Compared to ERL, RRI focuses on specific routes, compartments and agents located in those locations. RRI can compare the level of safety between cases that did not necessarily lead to death based on the proximity and length of exposure to the hazard.

3.2. EvacuSafe: proposed framework for safety evaluation of building evacuations

The integration of BIM, fire simulation, crowd simulation, and the proposed risk indices can result in a comprehensive and accurate framework for safety evaluation of building evacuation scenarios. Fig. 1 illustrates the proposed framework of EvacuSafe as elaborated in the following six steps.

3.2.1. Step 1: BIM model

BIM holds a collection of geometric and semantic information related to the building elements. This information includes geometry of the building and compartments, building material, location of smoke detectors, heat detectors, sprinklers, openings, and ventilation system. BIM is used for three purposes in this framework. The first is to provide information to the fire simulation module to forecast the expansion pattern of the fire. The second purpose is to support the path identification algorithm and help detect the possible egress routes for each compartment. The third purpose is to serve as a foundation for the crowd simulation module using ABM. These functions are detailed next.

3.2.2. Step 2: fire simulation module

The fire simulation module, which models the behavior and the expansion pattern of the fire, needs several user-defined inputs depending on the simulation approach and requirements of the specific engine used for modeling. For example, in FDS, the geometry of the building, building materials, fire loads such as furniture, location of doors and windows, ventilation type, and locations of sprinklers, smoke, and heat detectors must be determined to build a valid model.

FDS is a CFD model for simulation of fire-generated fluid flow. The migration of the fire and its byproducts is solved numerically using the large eddy simulation form of the Navier-Stokes equations, which are suitable for the low-speed, thermally-driven flow of heat and smoke from the fire. Finite difference methods are used to approximate the partial derivatives of the conservation equations of momentum, energy and mass. The solutions are then updated in time on a three-dimensional rectilinear grid [28]. A comprehensive review of the governing equations of heat transfer, energy transfer, combustion, and momentum is out of the scope of this paper.

FDS is a certified simulator by NIST and has been broadly used in many validation and verification research works. Although it has been selected for use in this research, any fire simulator reporting the tenability conditions in time and space can be applied in the fire simulation module.

The accuracy of the fire model will affect the evacuation assessment and model reliability. The information is captured from a BIM model in IFC format, thereby eliminating entry errors and the discrepancies

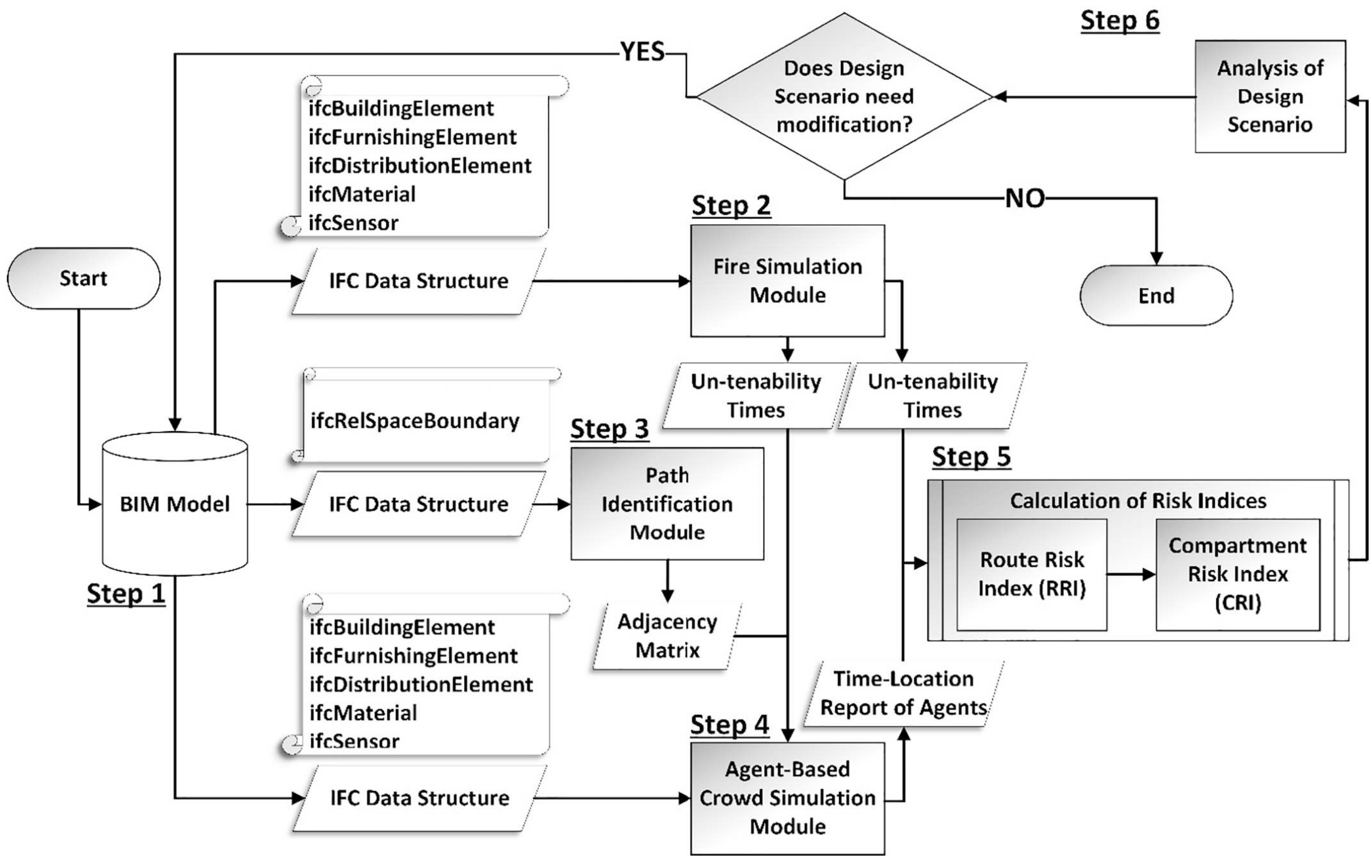

Fig. 1. Framework of EvacuSafe.

between the fire model and the building model. It also saves time and resources by avoiding making another building model in the fire simulator. Mapping the entities of *ifcBuildingElement* class in fire simulator builds the general geometry of the model. It includes *ifcShapeRepresentation* and *ifcLocalPlacement* classes that respectively report the shape and the placement of the element. *ifcFurnishingElement* transfers the data related to furniture and objects that can act as fire loads or obstacles in fire simulator. *ifcDistributionElement* holds information about distribution systems such as HVAC. *ifcSensor* is an indirect subset of *ifcDistributionElement*. It can be used for retrieval of information related to sensors such as heat, smoke or CO detectors.

The fire simulation outputs a report that documents the smoke distribution, temperature contours (isotherms), and poisonous gas dosage throughout the simulation period. Herein, un-tenability of each compartment can be defined based on the smoke alarm system. The smoke alarm is a notification for occupants before the conditions in the building become untenable. The fire simulation module must be able to generate a report of activation times for each detector. This report will be later used for calculations of ALET.

The equivalency of un-tenability and smoke detector activation time might be deemed too conservative. Research shows that smoke detectors normally provide enough time for occupants to escape before untenable thermal and toxic gas conditions develop [67]. However, the un-tenability criteria could be adjusted based on the requirements of each building.

3.2.3. Step 3: path identification module

The second connection is established between BIM and the path identification module. Herein, path identification is performed based on a network graph showing the interconnected structure of the navigable spaces in the building. In simple building plans, it may be possible to quickly generate the network graph manually. However, this task can take a significant amount of time and effort in multistory buildings with complex layouts. Therefore, an automated network graph generation via IFC data structures is proposed. This approach is applied so that it is not dependent on a specific BIM model file extension.

ifcRelSpaceBoundary was a new entity in IFC release 1.5 and it later became modified in release of 2X. This entity defines the relationship of a space to its surrounding elements, whether physical or virtual. *RelatingSpace* and *RelatedBuildingElement* are two attributes in *ifcRelSpaceBoundary* that explain the components of the connection. The two types of *RelatedBuildingElement* that matter here are *ifcDoor* and *IfcVirtualElement*. These two cases can determine navigable elements connecting two or more *ifcSpace* to each other. Fig. 2 illustrates the logic of constructing the network graph in this framework. In this terminology, a virtual element is a delimiter between the rooms or spaces. This element does not exist in the built environment and its only purpose is to allow decomposition of one space to smaller segments.

IFC file is parsed and all pairs of connections between the spaces, common doors and common virtual elements will be filtered. In this method, each *ifcRelatingSpace* and *ifcRelatedBuildingElement* (doors and virtual elements) are represented with nodes. The link between the nodes means entities are connected to each other. Size of each node is proportional to the number of links connected to that node. Fig. 3 schematically shows the network graph corresponding to Fig. 2. Representation of connections between the spaces with the help of graphs significantly helps the rapidity, transparency and comprehensibility of the presented information.

Fig. 3 is the visual representation of space/element connections in the building. The mathematical representation of navigation paths is normally presented in the form of an adjacency matrix. An *n-by-n* adjacency matrix, in which n is the total number of elements and spaces are filled with 0 and 1 values, indicates which pairs of nodes are adjacent. The first step towards calculation of RRI is finding all possible

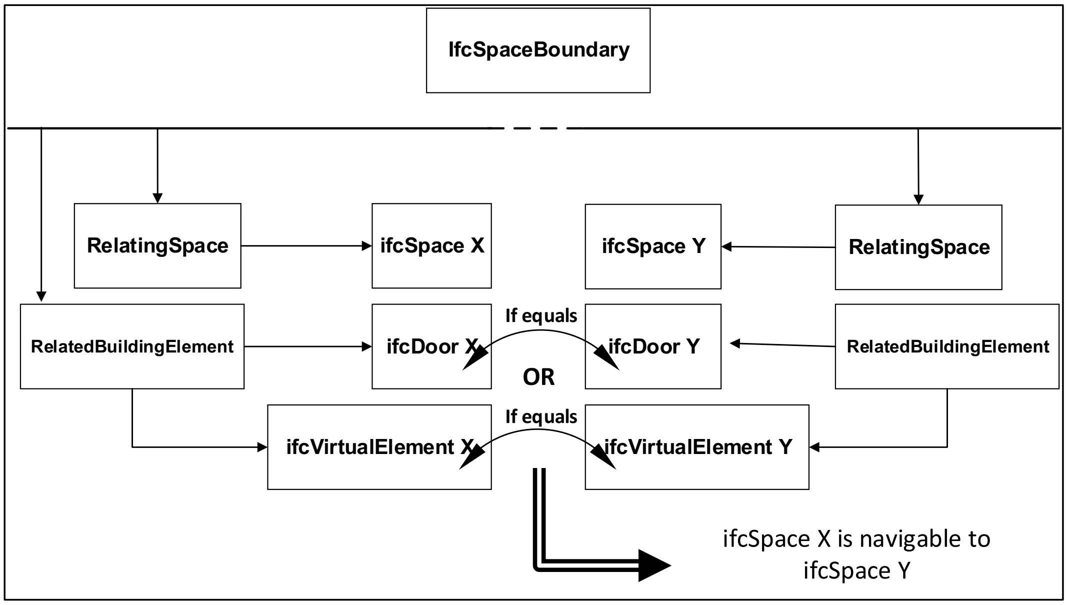

Fig. 2. Construction procedure of the network graph.

and separate egress routes for a compartment. To this end, a recursive code is used to scan the adjacency matrix and output the possible routes. The recursive code runs for each compartment and exhaustively searches in the adjacency matrix to find the connected nodes. In each run, one compartment is selected as the parent node and the matrix is explored for connected child nodes. At the end of the first iteration, all generated routes are saved in separate arrays. In the next iteration, child nodes are considered as new parent nodes and the code searches for next level child nodes. Generated routes will be again saved in new arrays. This continues until all of the routes reach exit doors. In this manner, all the possible routes leaving one compartment are achieved. Afterwards, the code runs for the next compartment until all the possible routes for all the compartments in the building are generated. The resulting routes are saved in arrays that convey the evacuees in their trips towards the exit doors. These arrays will be used in the next step of model development and will be explained further.

3.2.4. Step 4: agent-based crowd simulation module

The third link is between the BIM and the crowd simulation module. The ability of agent-based simulation in modeling populations, individual behaviors, and their interactions makes it the best alternative for simulating evacuating occupants. The adjacency matrix transfers the list of feasible egress routes from each compartment from the path identification module to the crowd simulation environment. Two categories of agents are created in the simulation. "Obedient" agents start from the target compartment and follow the assigned routes generated by the path identification module. "Unbound" agents are distributed on each floor of the building and are free to select a route for evacuation based on the parameters defined in the model and responses to environment and other agents. Obedient agents are placed and assigned such that there is one agent per egress route for every compartment. The importance of obedient agents is that they can be bound to a specific route and consequently their time-location movement report allows us to evaluate that particular route. The total number of agents is determined based on the occupant loads recommended by local or

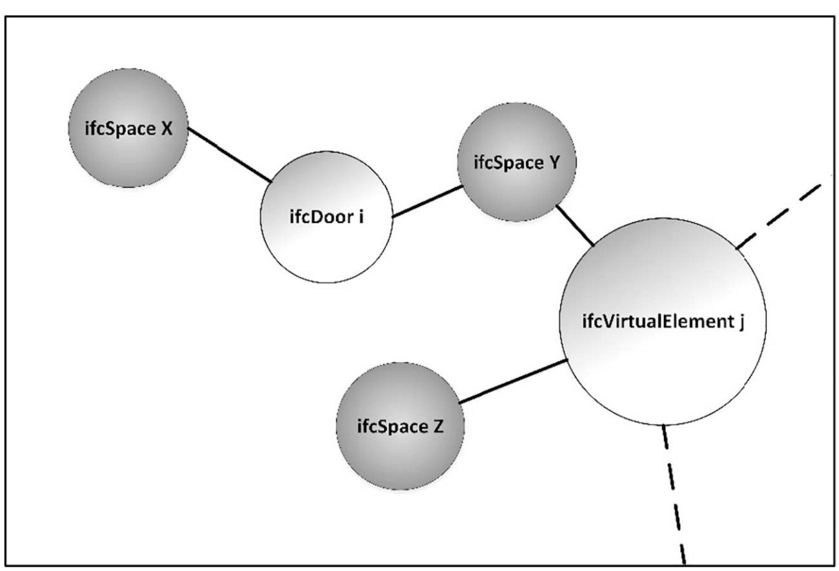

Fig. 3. Schematic representation of the network graph.

national building codes and regulations.

Crowd and evacuation simulation is performed by MassMotion© [68] from Oasys Software. The software is validated and verified for evacuation modeling [69]. In MassMotion, environment is represented as a continuous space and each agent can move freely over all coordinates. Route choice is at global level that is common in evacuation models and assumes that the agent knows the route from origin to destination [9,70]. The agent selects the route between the origin and destination according to the perceived cost of the routes among all possible routes. The cost of each route is based on three factors: distance, terrain type, and congestion [69].

$$Cost = \left(W_D \times \left(\frac{D_G}{V}\right)\right) + (W_Q \times Q) + (W_L \times L) \tag{4}$$

where:Cost = perceived total travel time along the route (s);W_D = “Distance” weight (agent property)(-);D_G = total distance from the agent position to the ultimate goal (m);V = desired velocity of the agent (agent property) (m/s);W_Q = “queue” weight (agent property) (-);Q = expected time in queue before reaching portal entrance (s);W_L = “geometric component traversal” weight (agent property) (-);L = geometric component type cost (s).

This cost perception is local, meaning that as an agent changes floors to approach the destination, a new local destination on the new floor is defined. The cost to the new local destination is calculated and the lowest cost is selected (Zarnke, cited in Mashhadawi [71]). This means that the agent will have no idea about the next floor congestion and terrain type. This emulation is supported as we see in the buildings it is recommended for evacuating via the nearest exit door on the floor [72], whereas the personal perception of an evacuee upon seeing a congested portal can change the direction. Free movement of the agent is also directed and restricted by a set of social forces. There are different forces on the agent that represent its motivation for movement towards destination. These forces combine in a resultant force demonstrating the direction and acceleration/deceleration of the agent. MassMotion accounts for several components of social forces including destination, neighbor, drift, queuing and corner. The destination force is the main element attracting the agent towards destination. Neighbor forces push the agents apart when they have different targets and are facing each other and pushes them towards each other when they have the same destination. The drift force is the directional bias for the cases where agents approach oncoming agents in the opposite direction. Queuing forces push the agents to the middle of the target, resulting in orderly queues. Finally, the corner force pushes the agent to have a wider turning radius on corners. In this setting, agents dynamically adjust their speed and direction based on the environment and their interactions. The details of the simulation approach can be found in Kinsey [69].

In each run of the evacuation simulation, one compartment is selected for detailed study. Obedient agents are assigned to that compartment, where the number of obedient agents is equal to the number of egress routes for that compartment. Unbound agents are evenly distributed throughout the building. The sum of the two types of agents is set to maximize the occupant load advised by building codes and other safety regulations. Unbound agents are free to seek any building exit door based on their perceived cost of each route, their interactions with environment, their interactions with other agents, and their obedience from social forces. Obedient agents, on the other hand, must follow the routes generated by the path identification module in the form of arrays of spaces and portals, which are introduced to MassMotion as waypoints. Obedient agents are constrained to follow these waypoints in their journey from the origin compartment to the building exit, but act like unbound agents between waypoints. Since obedient agents follow separate routes and are evenly distributed, as long as their ratio to unbound agents is negligible, they have little effect on the stochastic nature of the simulation. The simulation is repeated for each compartment. In each step, time-location reports about the obedient agents are exported.

Combining the un-tenability times from the fire simulator with the time-location data of the obedient agents will result in ALET values. The available safe egress time for an agent in a specific location is dependent on the un-tenability time of that location and the time when the agent is present in that specific location. Therefore, as opposed to other safety indices that are calculated using static formulas, the value of ALET is obtained from the simulation results within the EvacuSafe framework, which incorporates both fire and crowd simulation elements.

The stochastic movement of unbound agents keeps a level of uncertainty in calculating the travel time for each egress route. This randomness could be in terms of the varied speed of the agents, their directional bias when facing opposing flow, and queuing and congestion tolerances. This approach considers the effect of congestion in the calculation of RRI. Congestion will result in longer travel times. Likewise, congestion will result in slower escape speed and consequently a lower ALET at each point along the route. Both of these consequences increase the value of RRI for the route under study.

The last step is to incorporate the effect of fire evidence in movements and behavior of the occupants. In real-life scenarios, evacuees tend to avoid spaces where they sense smoke or other evidence of fire. Thus, the list of un-tenability times for the spaces are imported to the crowd simulation environment. As soon as each space becomes untenable, it will turn into a one-way inside-out zone. This means that the agents inside the untenable zone are allowed to exit, and no agent can enter the zone after un-tenability is triggered. The modeling can be improved and the speed and preferences of the obedient agents can be modified based on the effect of fire evidence on their behavior and mobility. This effect may be a result of exposure to high temperature or inhalation of poisonous fire products. On the other hand, fire propagation patterns could be affected by people's movement and actions. The ultimate goal to achieve the ideal connection and interaction between the fire simulation module and the crowd simulation module is to develop a unified simulation environment capable of simultaneously modeling both fire and crowd movement. The developed model needs to be able to perform high-accuracy, validated and verified simulation in both disciplines. In this way, the fire itself will act as an agent interacting with evacuating agents and building environment. Development of this unified simulation environment is outside the scope of this paper.

3.2.5. Steps 5 & 6: calculation of risk indices and analysis of design scenario

By integrating the results of the fire and crowd simulation modules, ALET values are calculated along the egress routes. RRI and CRI are calculated as per Eqs. (2) and (3), respectively. Based on these measures, analyses can be performed. Egress routes with infinite RRI are untenable even before the evacuee would be out of the building. Infinite RRI means ALET = 0 exists somewhere on the egress route. A blocked compartment is a compartment in which all of the connected egress routes have an RRI equal to infinity. Part of the analysis is to locate such compartments. At the end, the design scenario will be assessed based on the safety and availability of the compartments and egress routes under various fire scenarios. If any corrective modification is required, it will be applied to the BIM model. The aforementioned set of processes will iterate until a satisfactory design scenario is achieved.

4. Implementation & discussion

A two-story office building is chosen for implementation of the proposed framework. Fig. 4 shows 1st floor (a) and 2nd floor (b) plan views extracted from the BIM model. The building has three staircases, two of which are connected to outdoor space and the third connects the two floors together. The first floor has three exit doors that could be

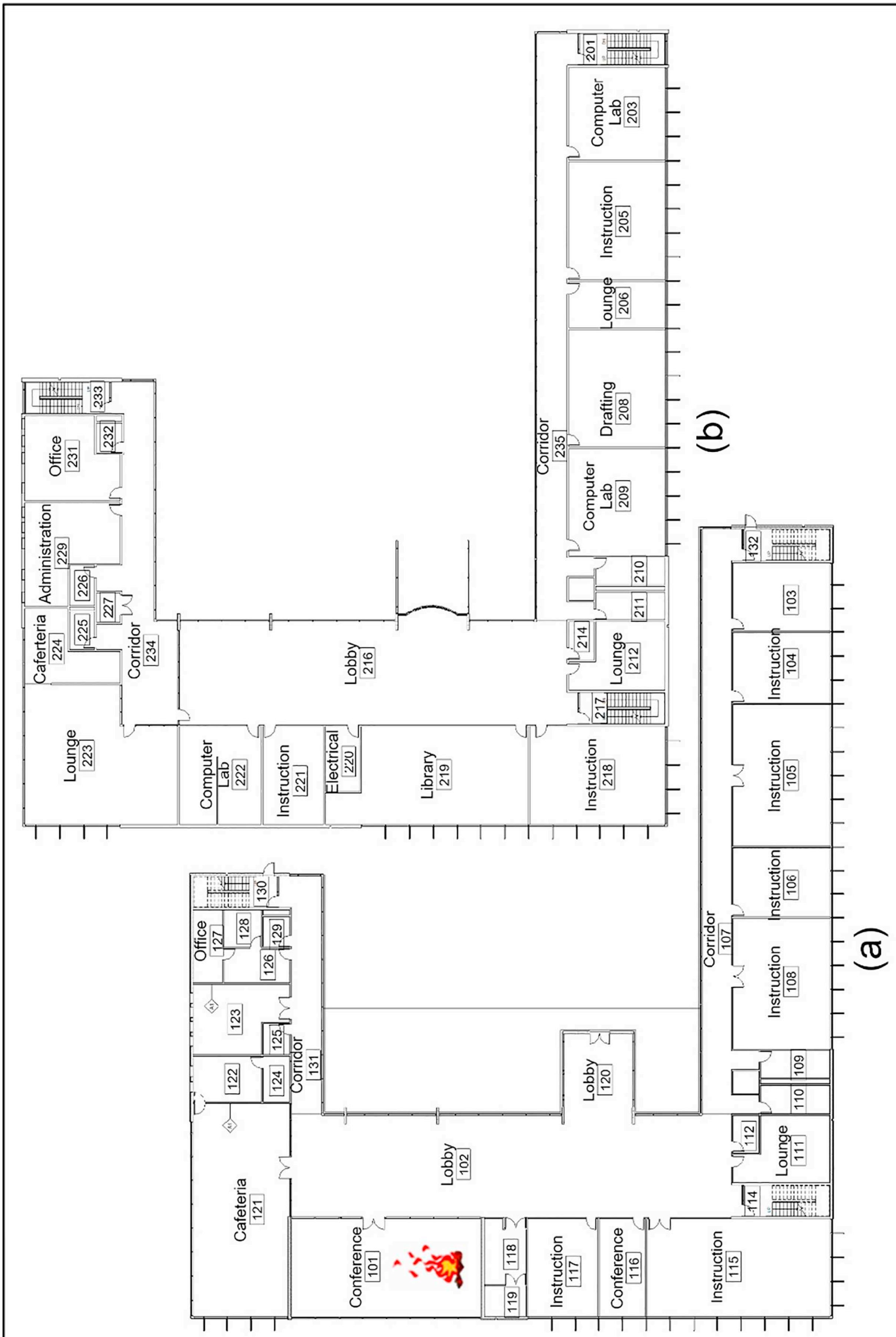

Fig. 4. Plan views of the first (a) and the second (b) floors.

used during emergency evacuation. The building's rooms, corridors, and staircases are divided into 59 compartments. Photoelectric smoke detectors and heat detector are installed in every compartment. Therefore, as per Step 1, the BIM model is captured and prepared to feed into the connected modules.

In Step 2, the model is loaded into a fire simulation software to test the building design in different fire scenarios. Herein, the model is imported into CYPECAD MEP [65] for fire analysis. The FDS tab of CYPECAD MEP performs the fire modeling in buildings based on two external modules. FDS carries out the analysis and SmokeView (SMV) visualizes the graphics. FDS and SMV do not have a user-friendly graphical interface; hence, CYPECAD MEP is more suitable for the analysis in the developed framework. IFC files can be uploaded into CYPECAD MEP, allowing a direct transfer of BIM into the fire simulation engine. This improves time efficiency and consistency in data transfer, promoting an automatic seamless connection between BIM and fire simulation.

The IFC file contains geometrical properties, materials, the location of smoke/heat detectors, and potential furniture as fire loads. In this example, it is assumed that the fire was initiated in compartment 101 located in the northwest corner of the first floor. This compartment is a lounge with a kitchen. This is based on reports that cooking is the leading cause of fire in residential and non-residential buildings [1,2]. The fire-initiating element is a 1 cubic meter block of wood representing furniture. Smoke and heat detectors are activated in 11% visibility obstruction (0.05 OD/m) and 57^0C respectively. In this research, the presence of smoke for visibility obstruction of 11% constitutes the early alert of an approaching fire. This is a conservative assumption and provides a larger safety margin to the results of analysis.

In Step 3, the IFC file is parsed to its entities. *IfcRelSpaceBoundary* class is captured and all of its instances are listed. The list is later filtered on the instances in which *ifcRelatedBuildingElement* is equal to *ifcDoor* or *ifcVirtualElement*. *ifcVirtualElement* could be the representation of an opening or a room separator. By assuming each building element or building space as a vertex in a network graph, an adjacency matrix is produced showing the dependencies between the spaces and the doors or virtual separators. The 3D model resulted in a 121 × 121 adjacency matrix that can be visually represented by the network graph shown in Fig. 5. Light gray vertices are spaces and darker ones are the doors or virtual elements. Nodes starting with S are regular room spaces, SC are corridors and SS are stairs. Nodes starting with D are doors, DE are building exit doors and V are virtual separators.

In Step 4, an ABM environment, MassMotion© [68] by Oasys Software, is used to perform the crowd simulation. MassMotion© is a software verified and validated based on NIST technical note 1822, which documents the process of verification and validation of building fire evacuation models [73]. This software enables the user to import the IFC file and transfer geometrical attributes of the model to crowd simulation environment. In MassMotion©, *ifcWall*, *ifcColumn* and *ifcFurnishingElement* are translated into barriers. *ifcSlab* is used to represent floors and *ifcStairFlight* shows stairs. The Ontario building code recommends 9.3 m^2/person occupant load for office spaces and 3.7 m^2/person for corridors [74]. The total area of both floors is 2920 m^2 comprising 367 m^2 of corridors and 2553 m^2 of offices, resulting in 99 agents in corridors and 275 in offices; this is rounded up to 400 agents distributed in these two floors. None of the codes suggests a distribution for the occupant load given that it is hard to foresee the location of occupants within a compartment. None of the codes suggest a firm pre-movement time, which is the interval between the first fire warning and agents' movements towards exits. In this research, pre-movement time is set to 20 s. It is assumed that the occupants are already trained for similar situations and can be quite responsive to warnings. In addition, it is an office building and mostly occupied during the day, therefore occupants are relatively alert to emergencies. The 20 s assumption is within the range of 10–55 s, 12–105 s and 5–173 s observed in the monitoring of evacuees in office buildings [75–77]. Agents are represented by a radius of 0.25 m. A body ellipse of 0.6 m by 0.4 m was recommended [78], which was translated into a body circle with the radius of 0.25 m to result in the same area while lowering the computational load in agent movements and interactions [69]. The speed of agents is normally distributed with a mean of 1.35 m/s and standard deviation of 0.25 m/s. It is assumed that the speed does not go above 2.05 m/s or lower than 0.65 m/s. This distribution is based on monitoring of commuters with different genders and ages [78].

The adjacency matrix resulting from the *ifcRelSpaceBoundary* class has all of the possible routes between each compartment and the exit doors. A recursive code in MATLAB [79] has been applied to the adjacency matrix so that all possible egress routes pertaining to each compartment are produced in the form of arrays. The first element of each array holds the compartment under study and the last element is the exit door. The arrays are imported to MassMotion© as optional routes for the obedient agents. For example, consider that compartment 212 on the second floor has 9 possible egress routes. Nine obedient agents begin their journey from compartment 212 with each taking a different path to exit. Considering that 9 obedient agents constitute only 2.25% of the total agents in simulation, the ratio is low enough (0.023), and it will not impact the presumed un-navigated environment of the evacuation, surrounding the obedient agents. The number of obedient agents used in the simulation does not exceed 9, as this is the maximum number of possible routes for the second floor compartments. It is worth noting that having 9 obedient agents leaves 391 (400–9) unbound agents in the simulation.

The presence of fire within the evacuation scene can affect the behavior of occupants as they panic and change their routes. To embed this phenomenon in the modeling process, some gates are located at the portals of the compartments. Device activation times from FDS are imported as a timetable to schedule open/close actions of the gates. Therefore, when the un-tenability condition prevails, the zone turns into an exit-only zone i.e. no agents can enter it. As mentioned in the model development section, the realistic interactions of fire and evacuees are definitely more than the escape response of evacuees to smoke. However, the limitations of simulation software applications affect the implementation process. FDS and MassMotion are designed for different purposes and they do not accommodate real-time data transfer between themselves. This is a limitation for the implementation of this framework; however, understanding all the aspects of this limitation makes a significant opportunity for further research on development of a validated unified simulation software answering to both needs. This software would have to be: 1) validated and verified compliant with evacuation and fire safety codes; 2) able to model large-scale facilities; 3) capable of importing IFC data structure; 4) able to include the effect of fire on mobility and behavior of the agents; 5) able to consider the impact of evacuees' movements and actions on fire growth patterns (e.g. effect of opening/closing doors on smoke propagation).

In Step 5, the outcomes of the aforementioned modules are combined to calculate the risk indices. The time of entry to each compartment of each agent and its total travel time to the exit is recorded. The combination of these times and the device activation times will generate the RRI for each route. ALET for each compartment is the device activation time minus the elapsed agent time of travel from the origin compartment to the compartment under study. If the value is negative, it means that smoke reached that point before the agent arrived. In the analysis, negative values are substituted by 0. Fig. 6 relates to compartment 212 with 3 safe egress routes out of 9 possible routes, and shows 1/ALET vs. elapsed time. The remaining 6 routes were blocked due to the presence of smoke during the evacuation. To ensure a margin of safety, the routes that indicate that an agent will get closer to the smoke within 10 s are eliminated. As per Fig. 6, agent 7402 at elapsed time 0:01:01 was closest to un-tenability, with 1/ALET = 0.09. RRI (area under the 1/ALET curve) of routes 212-7401, 212-7402 and 212-7403 are 0.438, 1.385 and 0.276, respectively. Based on our desire to

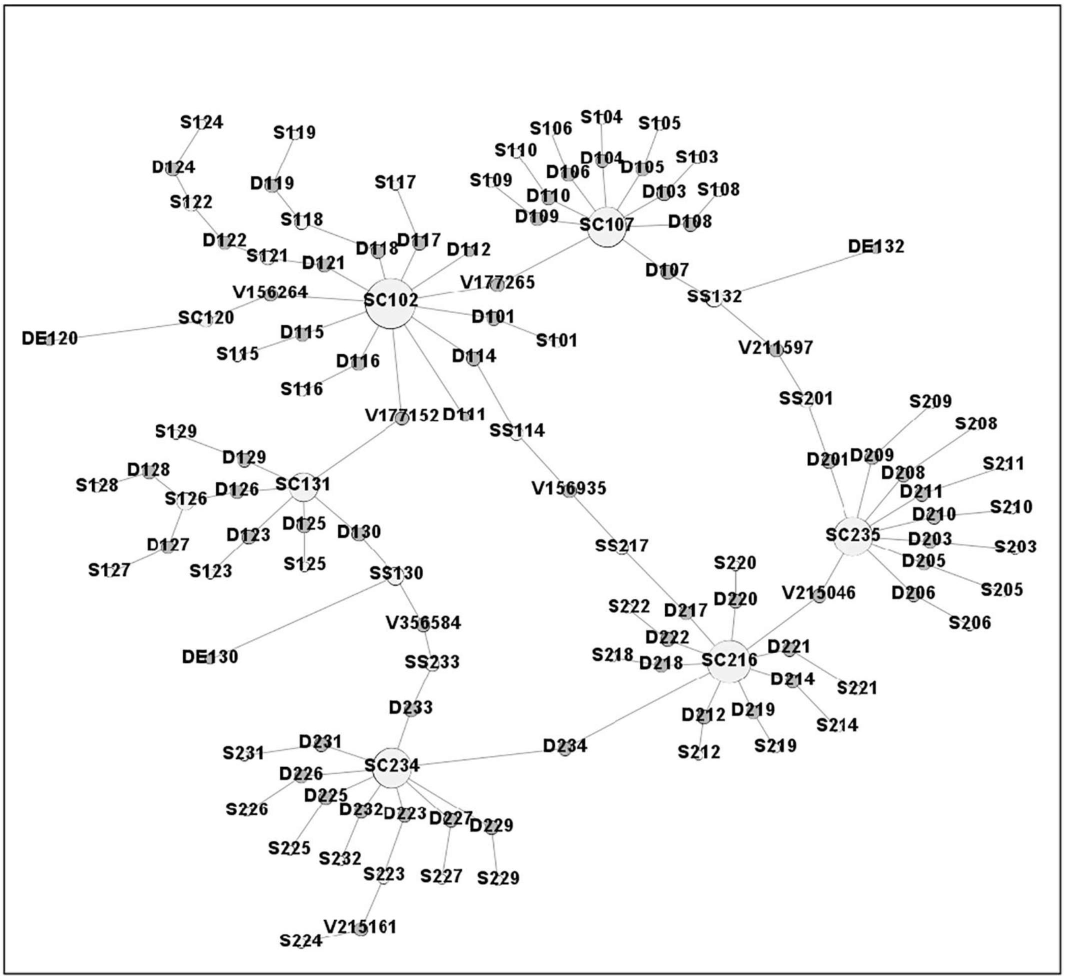

Fig. 5. Network graph of the building.

minimize risk and therefore RRI, route 7403 is preferred. This plot shows how a shorter route like 212-7402 is not necessarily safer among the routes exiting the same compartment. The analysis shows that for many compartments including 209 and 211, agents taking the shortest route encounter smoke in the first floor corridor, making their RRI equal to infinity.

Given that the formula of RRI is an integral over time, any change in the schedule of evacuation will be tracked. This has two benefits. First, for a particular point, the decision to evacuate or stay-in-place could be compared. It was previously mentioned that the elapsed time begins even before the agent starts to escape. In the stay-in-place scenario, the agent is stationary and its relative distance to the fire changes the ALET value, and consequently the area under the curve of 1/ALET vs. elapsed time (i.e. RRI). Second, staged evacuation strategies can be investigated

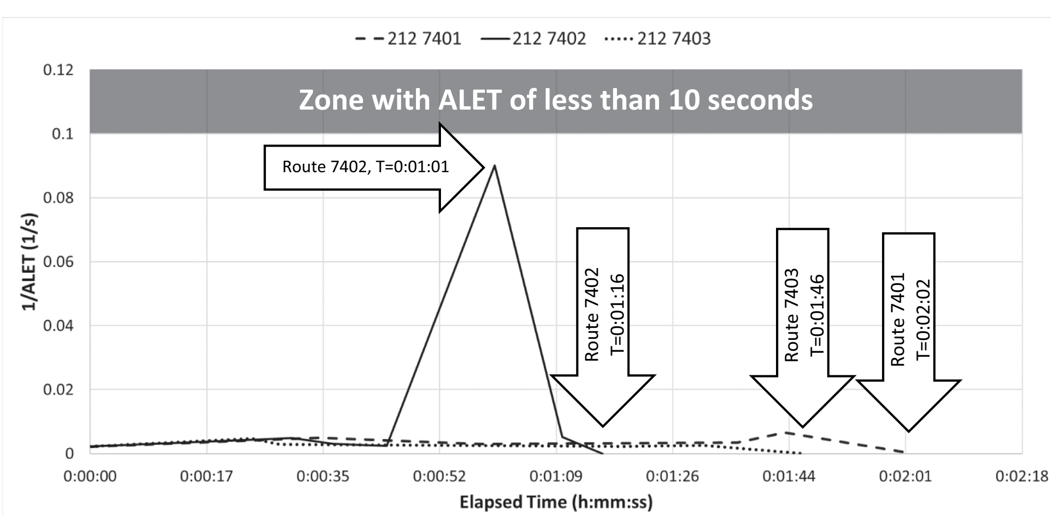

Fig. 6. 1/ALET vs. elapsed time.

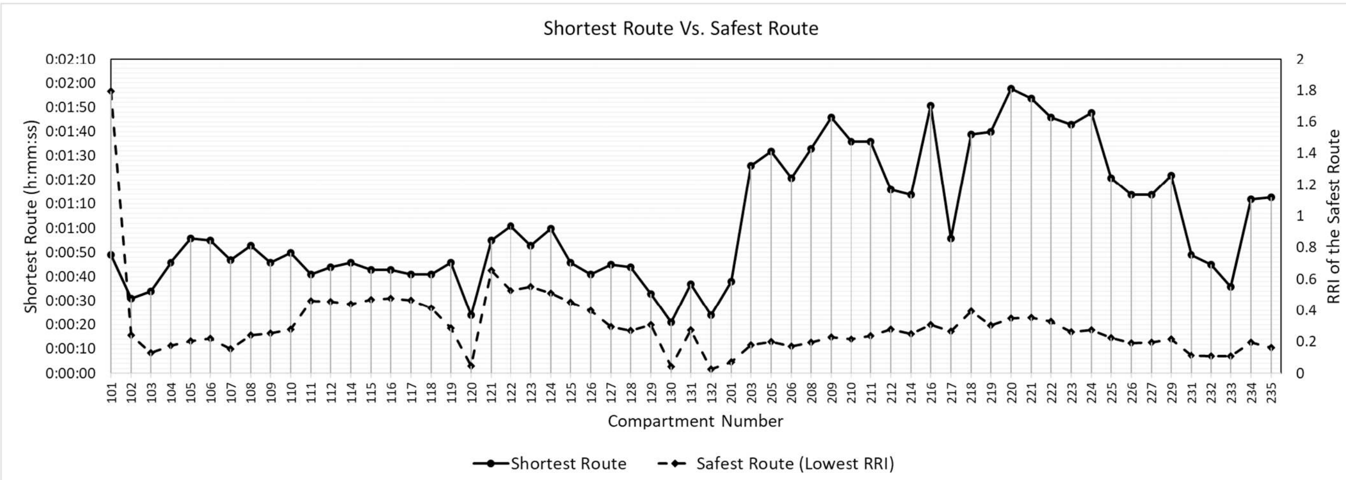

Fig. 7. Shortest route vs. safest route.

to determine the impact of evacuation delays on RRI and CRI.

The shape of the 1/ALET vs. elapsed time curve is also important. For a specific RRI value, which means equal areas under the curve, an egress route with a constantly descending slope is preferred to one with peak-shaped portions. A strictly descending plot indicates that the agent is always moving further from the hazard along the egress route.

Conversely, Fig. 7 shows the shortest route and the lowest RRI by compartment for the example fire scenario. The purpose of this figure is to compare how RRI and the shortest routes vary by compartment, and these measures are not necessarily from the same egress route. The upper solid line represents the time to complete the shortest route as measured on the left vertical axis. The lowest RRI associated with the compartment is the lower dashed line and is scaled to the right vertical axis.

As expected, compartments on the second floor (200 s compartment tags) have longer exit routes; however, their lowest RRIs are often lower than the RRIs in the first floor. Because the fire is initiated in the first floor, the agents in the second floor have less exposure to it. Consider compartments 105 and 111; while 111 has a shorter exit path, its best RRI is higher than the best RRI for compartment 105 due to its proximity to the fire. Not surprisingly, the correlation between these two measures is very low ($R^2 = 6.2E{-}4$), indicating that they provide valuable and independent insights to compartment egress safety. Fig. 7 supports the hypothesis that a compartment closer to an exit is not necessarily safer.

Fig. 8 shows the utilization (black bars) and blockage rates (gray bars) of each compartment. The utilization rate is the percent of exit routes that go through a compartment based on the network graph. The blockage rate is the percent of simulations in which a compartment is blocked by smoke. While compartments 102 and 131 are highly used by evacuees, they are blocked at least half the time that they are used (Fig. 8a). Fig. 8b shows the utilization and blockage rates when the occupants take the safest route (lowest RRI). By taking the safest routes, the utilization rates are lowered for compartments 102 and 131; and blockages are eliminated.

The crowd simulation was run in two modes, both of which had the fire start in compartment 101 as before. In the first, agents randomly take an exit route without considering their safety. In the second, agents are forced to take the route with the lowest RRI. The total evacuation time for the first and the second modes are 2:25 and 3:21 respectively. The safest routes are shown in Fig. 9. In the second mode, agents on the second floor are prohibited from staircase compartment 217, shown as the decreased utilization rate of compartment 217 from Fig. 8(a) to (b). Thus, in the second mode the total evacuation time is longer but the safety of the occupants is secured by deviating their exit routes towards the compartments that are farther from the fire.

Scrutinizing the avoidance of compartment 217 provides a clear understanding and vision about the application of RRI. The results of simulation show that all the routes passing through compartment 217 are prohibited because they would be later blocked in corridor 102. Untenability condition prevailed at $t = 81.4$(s) in compartment 102. The closest second floor compartment to staircase 217 is compartment 218. The fastest obedient agent from compartment 218 arrives in compartment 102 at the time of $t = 106$ (s). In this way, occupants from the second floor are not fast enough to pass compartment 102 before untenability condition prevails. However, for a first floor compartment such as 117, the slowest agent passes corridor 102 at $t = 48$ (s). Therefore, the blockage of one route in one compartment depends on the origin of that route. In this way, the effectiveness of RRI is demonstrated by tracking the status of the agent all along its evacuation travel.

As per the results of the simulation in Fig. 9, the optimal route for compartment 121 is to Exit 2. Visually, it seems that the route to Exit 1 is shorter and is even further from the source of fire. Travel distances of 27 (m) and 34 (m) for exits 1 and 2, respectively, confirms this estimation. However, crowd simulation shows that the agent exits from Exit 1 in $t = 76$ (s) and from Exit 2 in $t = 55$ (s). Thus, the travel time for Exit 1 is 21 s longer. This is because of congestion near Exit 1. The second floor occupants primarily use Exit 1 and Exit 3. Therefore, Exit 2 is only used by central compartments of the first floor, and consequently is less congested. This shows how optimal route selection includes the effect of congestion on RRI calculations.

Interestingly, Fig. 9 shows how adjacent compartments produce clusters of safe egress routes. Consider that the building is divided to several zones such that compartments in each zone have similar exit routes towards the same staircase or exit door. Recently, an emerging concept called dynamic exit signage has been proposed [80]. The exit sign alters its direction based on the instructions from a human operator or expert system. Therefore, the proposed system can help in finding clustered zones and locating the best points to install the dynamic signs. Defining the exact location of the dynamic signs is a challenging study and out of the scope of this paper. There are situations where the safest routes for two adjacent compartment portals direct people to different directions. Compartments 112 and 110 exiting at the border of corridors 102 and 107 are good example for such situations. In practice, it is barely possible to divide the occupants crossing the same area based on their compartments of origin. Therefore, placement of the exit signs would be a trade-off between decreasing the accepted tolerance in choosing the lowest RRI and not confusing the evacuees about egress paths.

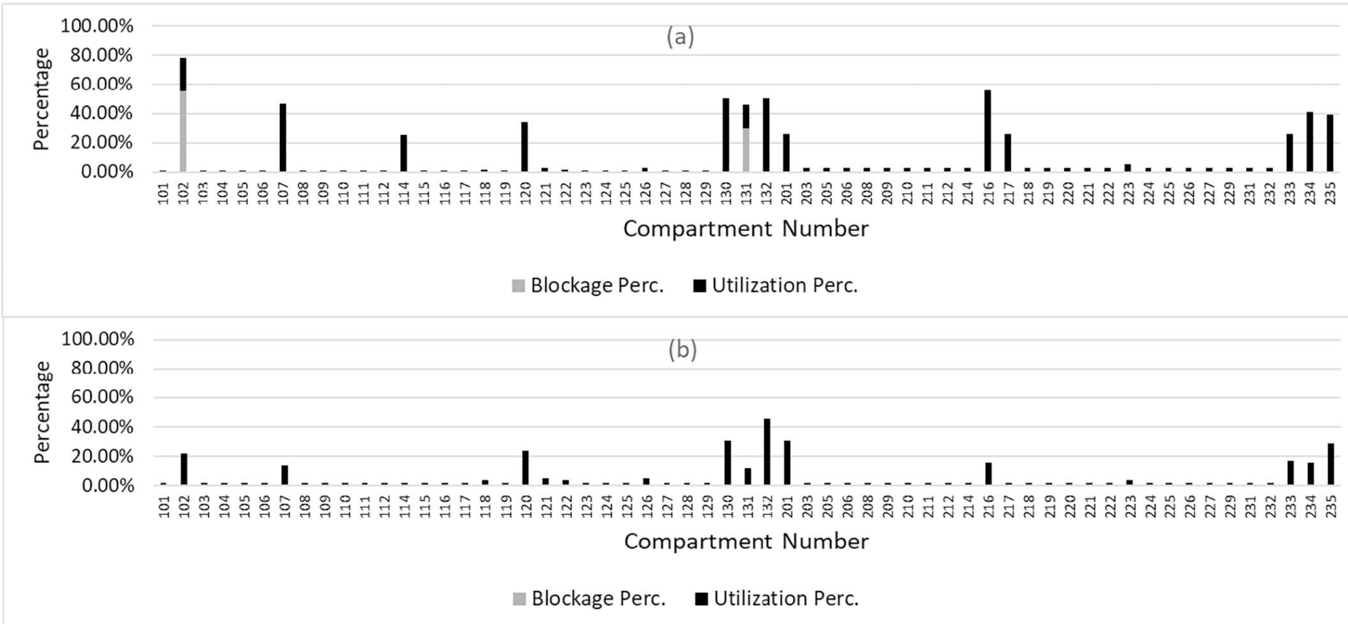

Fig. 8. Utilization/blockage percentage of compartments before (a) and after (b) safest route analysis.

Table 1 tabulates RRI, CRI, and number of available routes for each compartment. The color code is based on equivalency to a 220-second trip (0:03:40) with different ranges of ALET, categorized to 4 zones. This 220-second trip is chosen since it is the longest travel time observed from the crowd simulation. Therefore, this classification puts the analysis on the conservative side. Table 1 can help designers evaluate the performance of each compartment during evacuation. Detection of the safest compartment is the result of a multi-objective optimization that tries to find the compartment with maximum number of possible routes and minimum value of CRI. The decision on which of these two factors are more important for evaluation depends on expert opinion and the characteristics of the design scenario, which is beyond the scope of this paper.

Knowing that the value of RRI is dependent on travel time, proximity to fire, and the length of exposure, the scale of RRI will vary with scenario. For example, the maximum travel time in this building is 220 (s) and the minimum acceptable ALET is 10 s; therefore the maximum RRI must be 12 to have a safe route. This scale is dependent on several factors including the building characteristics, pre-defined safety thresholds, and human factors. The RRI is defined to compare the safety

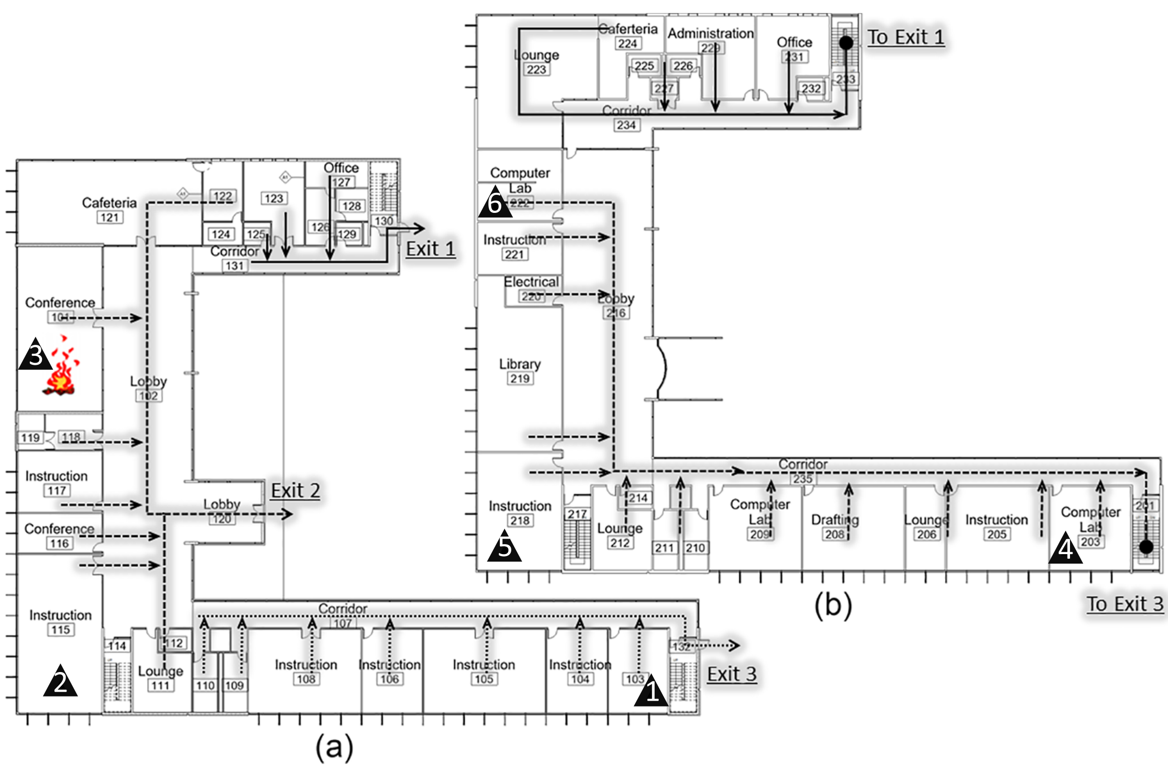

Fig. 9. Route map after safest route analysis.

Table 1
RRIs, CRI and number of available routes.

Comp. Tag	Comp. Type	RRI					CRI	Number of Available Routes
		Route 1	Route 2	Route 3	Route 4	Route 5		
101	Room	2.80	1.79	2.22			2.27	3
102	Corridor	1.27	0.24	0.56			0.69	3
103	Room	1.50	0.13				0.82	2
104	Room	0.91	0.18				0.54	2
105	Room	3.41	0.88	0.20			1.50	3
106	Room	2.20	0.54	0.22			0.99	3
107	Corridor	2.49	0.98	0.15			1.21	3
108	Room	2.22	0.44	0.24			0.96	3
109	Room	1.53	0.31	0.25			0.70	3
110	Room	1.67	0.32	0.28			0.76	3
111	Room	1.99	0.46	0.60			1.01	3
112	Room	2.16	0.46	0.53			1.05	3
114	Stair	1.59	0.44	0.50			0.84	3
115	Room	1.72	0.47	0.65			0.95	3
116	Room	1.75	0.48	0.70			0.98	3
117	Room	1.56	0.47	0.68			0.90	3
118	Room	1.29	0.41	0.69			0.80	3
119	Room	1.31	0.29	0.55			0.72	3
120	Corridor	1.58	0.04	0.62			0.75	3
121	Room	0.91	0.65	1.07			0.88	3
122	Room	0.88	0.53	1.06			0.82	3
123	Room	0.55	0.78	1.14			0.83	3
124	Room	1.05	0.51	1.22			0.93	3
125	Room	0.45	0.68	1.20			0.78	3
126	Room	0.40	0.86	1.38			0.88	3
127	Room	0.29	1.02	1.64			0.99	3
128	Room	0.27	0.88	1.81			0.98	3
129	Room	0.31	0.84	1.57			0.91	3
130	Stair	0.04	1.04	2.78			1.28	3
131	Corridor	0.28	0.80	1.22			0.77	3
132	Stair	1.93	0.02				0.98	2
201	Stair	0.48	0.07				0.27	2
203	Room	0.50	0.18				0.34	2
205	Room	0.45	0.20				0.32	2
206	Room	0.47	0.17				0.32	2
208	Room	0.40	0.19				0.30	2
209	Room	0.39	0.23				0.31	2
210	Room	0.42	0.22				0.32	2
211	Room	0.40	0.24				0.32	2
212	Room	0.44	1.39	0.28			0.70	3
214	Room	0.39	1.13	0.25			0.59	3
216	Room	0.37	0.31				0.34	2
217	Stair	0.43	0.39	0.27	2.38	0.69	0.83	5
218	Room	0.46	0.39				0.42	2
219	Room	0.44	0.30				0.37	2
220	Room	0.41	0.35				0.38	2
221	Room	0.39	0.35				0.37	2
222	Room	0.34	0.33				0.34	2
223	Room	0.26	0.33				0.29	2
224	Room	0.27	0.33				0.30	2
225	Room	0.22	0.34				0.28	2
226	Room	0.19	0.34				0.27	2
227	Room	0.19	0.33				0.26	2
229	Room	0.22	0.45				0.33	2
231	Room	0.11	0.40				0.26	2
232	Room	0.11	0.41				0.26	2
233	Stair	0.11	0.44				0.27	2
234	Room	0.19	0.38				0.29	2
235	Room	0.44	0.16				0.30	2
	ALET (s)	ALET>440	440>ALET>220	220>ALET>110	110>ALET			
	RRI Range/ Color Code	0-0.5	0.5-1	1-2	2-12			

Table 2
Result of 6 fire scenarios.

Scenario	Fire initiating compartment	Floor	Location	List of blocked compartments						Number of blocked compartments	RRI (MIN, MAX)	CRI (MIN, MAX)
1	103	1	SE	104	105	106	108	109	110	6	0.02, 3.41	0.16, 2.27
2	115	1	SW	122	124					2	0.03, 1.70	0.09, 1.32
3	101	1	NW							0	0.03, 3.91	0.03, 2.58
4	203	2	SE	205	206	208	209	210	211	6	0.00, 1.49	0.02, 1.37
5	218	2	SW	219						1	0.00, 4.69	0.04, 3.24
6	222	2	NW							0	0.00, 2.88	0.05, 2.45

of routes and compartments in the same building and not to other case studies. Likewise, travel time and distance are not informative and countable when comparing a high-rise building to a house, for example.

On occasion, people in adjacent rooms may be directed in opposite directions, which can be difficult in a real situation. For example, compartments 109 and 110 are at the edge of separation between routes to Exit 1 and Exit 2. These compartments are very close to compartments around corridor 102 and the evacuees might be confused to find the lowest RRI route. As per Table 1, both compartments show Route 3 going to Exit 3 as the lowest RRI. However, the RRI of Route 2 going to Exit 2 is in the same RRI zone as Route 3. The distinction between optimal and near-optimal is unclear, especially when it had logistical implications. Using near-optimal solutions allows evacuees the option of passing through corridor 102 to Exit 2, providing route selection flexibility in the same RRI zone.

To measure the impact of a fire in different locations of the building, 6 fire scenarios are developed in the EvacuSafe framework. The fire initiating point is moved to the northwest, southwest and southeast corners of the building in both floors, as indicated by black triangles in Fig. 9. The numbers on the black triangles represent the scenario numbers in Table 2. All assumptions regarding the agents' behavior and safety criteria remain the same. Table 2 summarizes the results. The highest number of blocked compartments is recorded in scenarios 1 and 4, both in the southeast end of the building but on different floors. The southwest and northwest corners of the building stand in second and third place. This shows that a fire in the southwest part of the building, on either floor, can lead to a higher risk of blockage and a more critical condition.

It is important to track blocked compartments when examining different fire scenarios. If a specific compartment is repeatedly blocked, it could trigger a design review of the compartment. In Table 2, none of the blocked compartments has appeared more than once. Therefore, none of the compartments appears to have an intrinsic design issue.

5. Conclusion

In this research, EvacuSafe was developed as a framework for evaluating the evacuation safety performance of buildings based on multi-criteria risk indices and through the integration of agent based modeling, fire simulation tools, and building information models. Specifically, two spatiotemporal risk indices, Route Risk Index (RRI) and Compartment Risk Index (CRI), were defined to quantify the safety of the egress routes and building compartments. Instead of using an overall building risk index, these indices allow for safety evaluation of individual compartments and routes within the building. Using these indices with integrated fire and crowd simulation tools, EvacuSafe was shown to detect blocked egress routes and compartments, find safest egress route from each compartment, and determine the most critical locations of fire initiation in the building for different fire scenarios.

The EvacuSafe results for six fire scenarios on a two-story building with 59 compartments demonstrated that the developed framework provides a more comprehensive evaluation of the evacuation safety parameters within the building compared to safety indices that are currently used in the industry. The IFC-based integration of the BIM model with fire simulation tools and ABM within the EvacuSafe framework, provides designers with the ability to analyze a building layout in BIM under various fire scenarios and ultimately allow for design optimization based on multiple safety criteria.

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